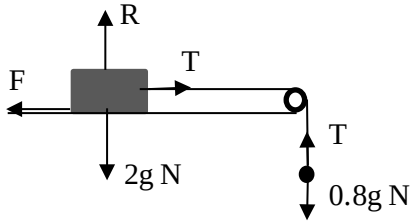
11	**(a)**			**B1** **B1** **B1** **[3]**	**1.1a** **1.1a** **1.1a**	both weights correct. common tension in the rigtht directions Friction and normal reaction and no extra forces	Allow weight of box and weight of sphere, but not if both marked weight. Allow T_1 and T_2 provided they are clearly shown equal to each other elsewhere. For F,allow $0.35R$, $0.35\times 2g$ $0.7g$ or 6.86 N
11	**(b)**		$T-F=2a$ $0.8g-T=0.8a$	**B1** **B1** **[2]**	**1.1a** **1.1a**	Allow any expression for F Allow distinct tensions if consistent with diagram	For F,allow $0.35R$, $0.35\times 2g$ $0.7g$ or 6.86 N
11	**(c)**		Vertically for the block $R=2g$ Friction $F=0.35R=0.7g$ Add equations $0.8g-F=2.8a$ $0.8g-0.7g=2.8a$ $a=0.35$ m s^{-2}	**M1** **A1** **M1** **A1** **[4]**	**3.1b** **2.1** **2.1** **2.1**	Attempt to use μ to evaluate friction Correct value for F Eliminate T from their equations AG must follow from correct work	Some of this work may already have been seen in previous part
11	**(d)**		Use $s=0.5,\quad u=0,\quad a=0.35$ $0.5=\frac{1}{2}\times 0.35\times t^2$ $t=1.69$ s	**M1** **A1** **[2]**	**1.1a** **1.1**	Using *suvat* equation(s) leading to a value for t Do not allow ± 1.69	